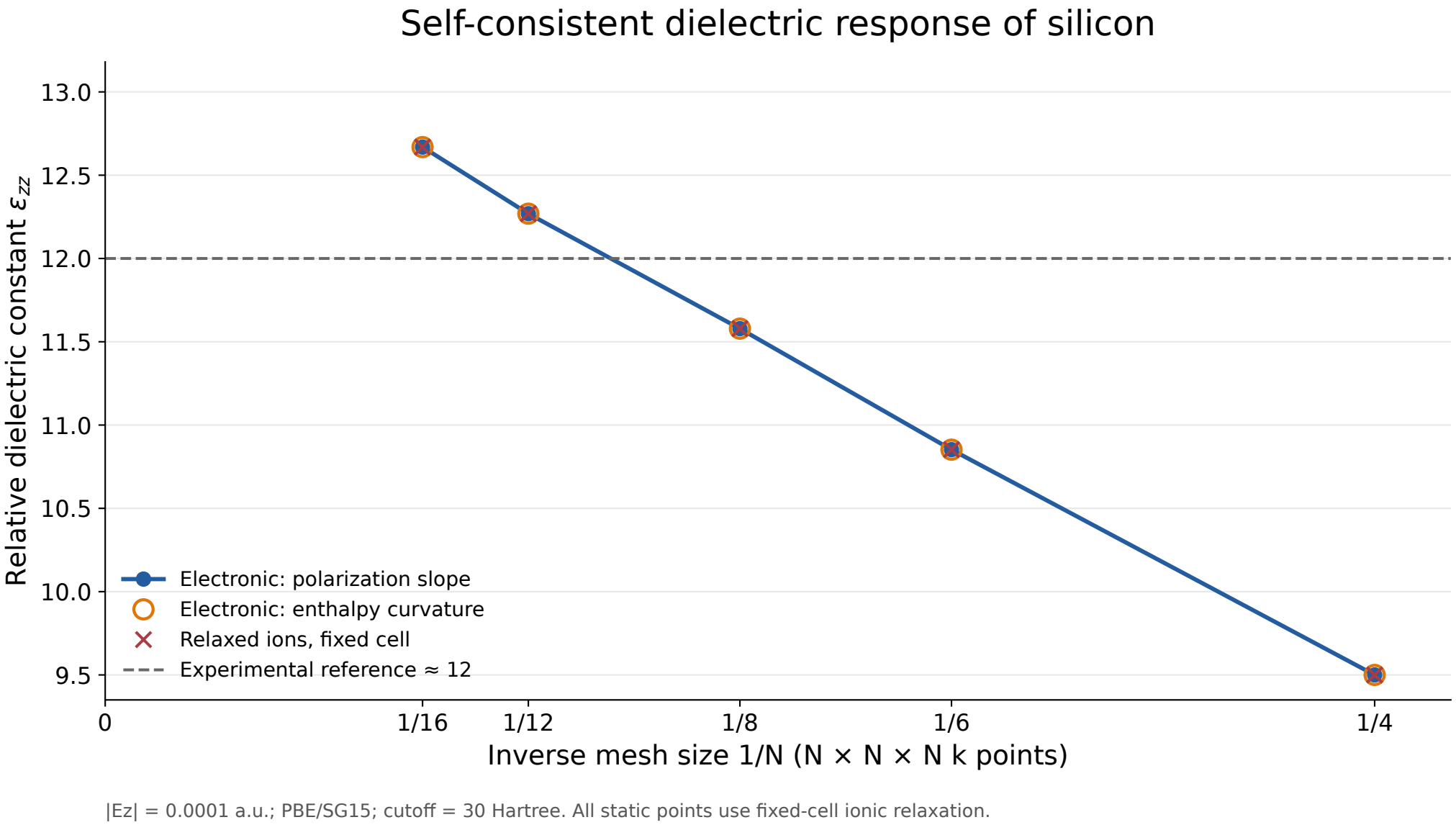
Self-consistent dielectric response of silicon
Relative dielectric constant $\epsilon_{zz}$
13.0
12.5
12.0
11.5
11.0
10.5
10.0
9.5
0
1/16
1/12
1/8
1/6
1/4
Inverse mesh size 1/N (N × N × N k points)
Electronic: polarization slope
Electronic: enthalpy curvature
Relaxed ions, fixed cell
Experimental reference ≈ 12
|Ez| = 0.0001 a.u.; PBE/SG15; cutoff = 30 Hartree. All static points use fixed-cell ionic relaxation.